Credit EDA Assignment

BY:

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Problem Statement:

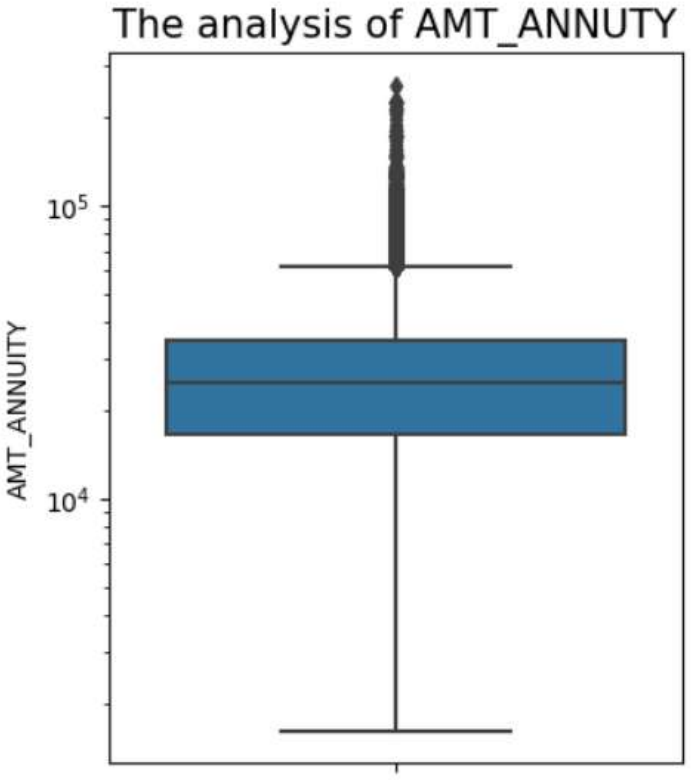
The aim of the EDA case study is to find out the patterns in the repayment of loans by clients based on the clients difficulty.

Based on the analysis, the Bank can deny the loan, reduce the loan amount, lending to the clients at a higher rate for the risky applicants, etc. The bank must not reject consumers who are capable of repaying the loan.

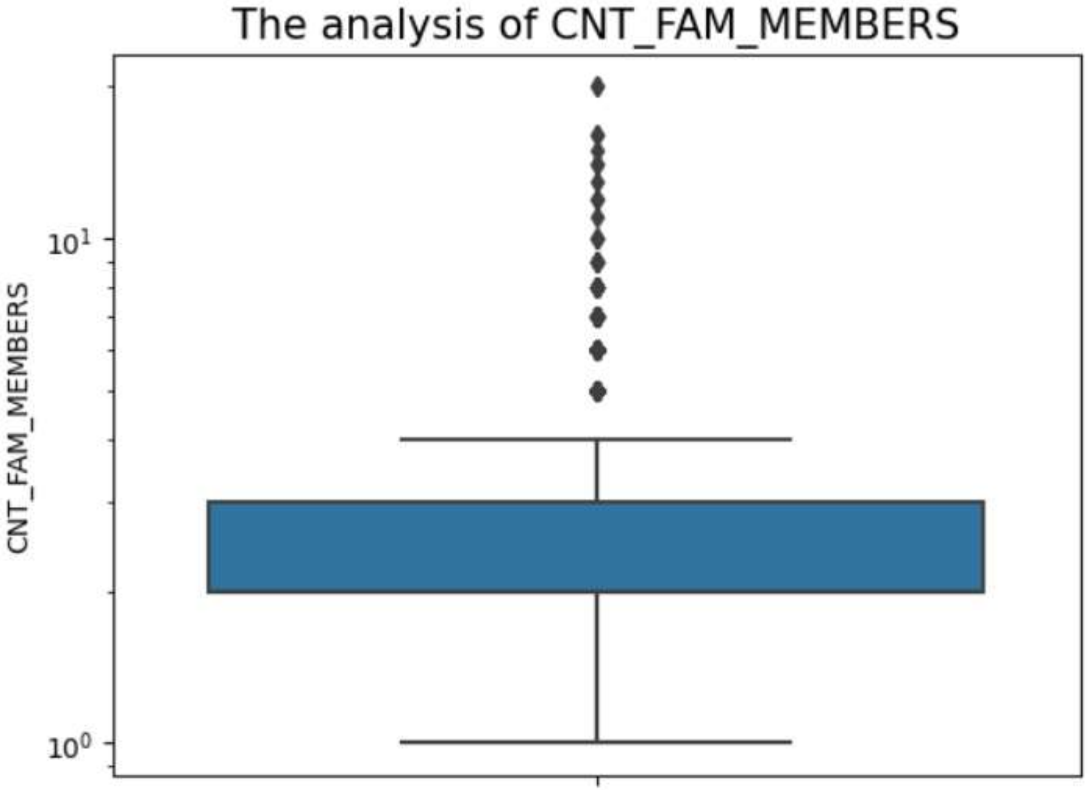
Approach:

- Data loading
- Data cleaning
- Data binning
- Univariate Analysis
- Bivariate Analysis, Correlation
- Merging of data
- Univariate and Bivariate Analysis

Analysis

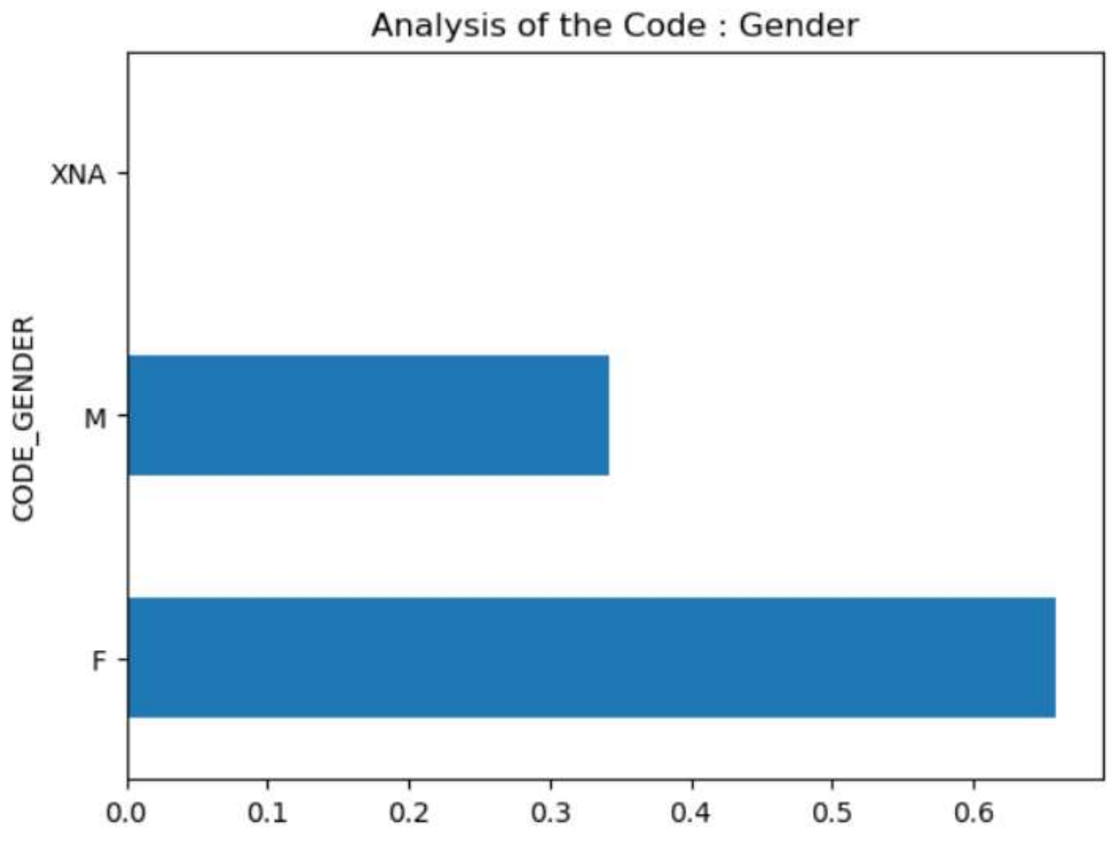


Inference: The Box Plot reveals the presence of several outliers, indicating a significant disparity between the maximum and minimum values. Therefore, we have decided to use the median value to replace the null values.

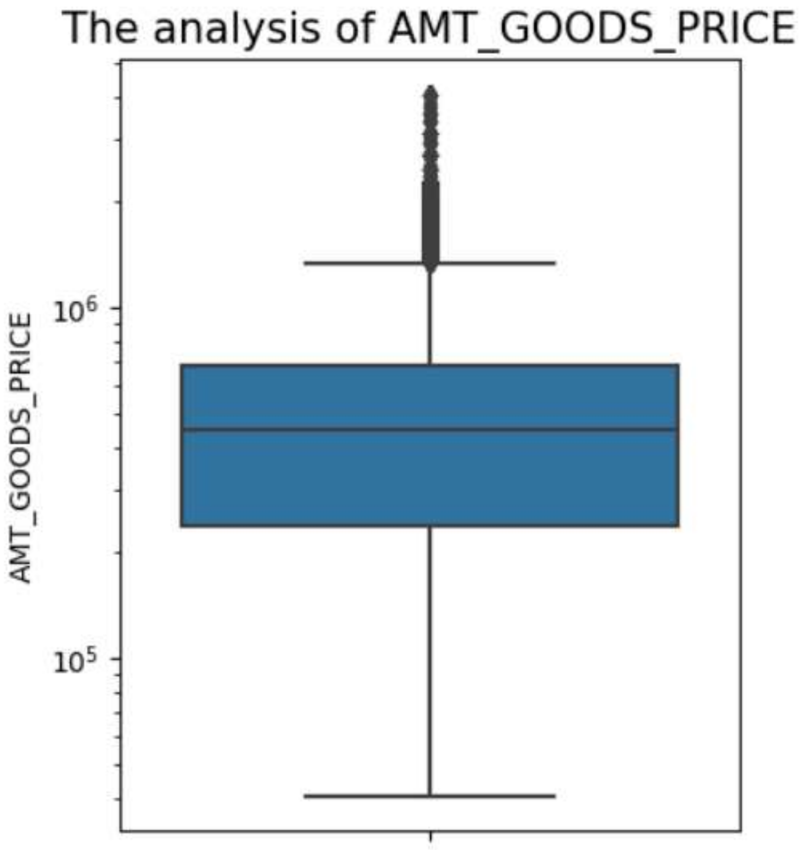


Inference: The Boxplot indicates the presence of significant outliers and a notable difference between the 75th percentile and the maximum value.
Consequently, we will use the median value to replace the null values.

Analysis

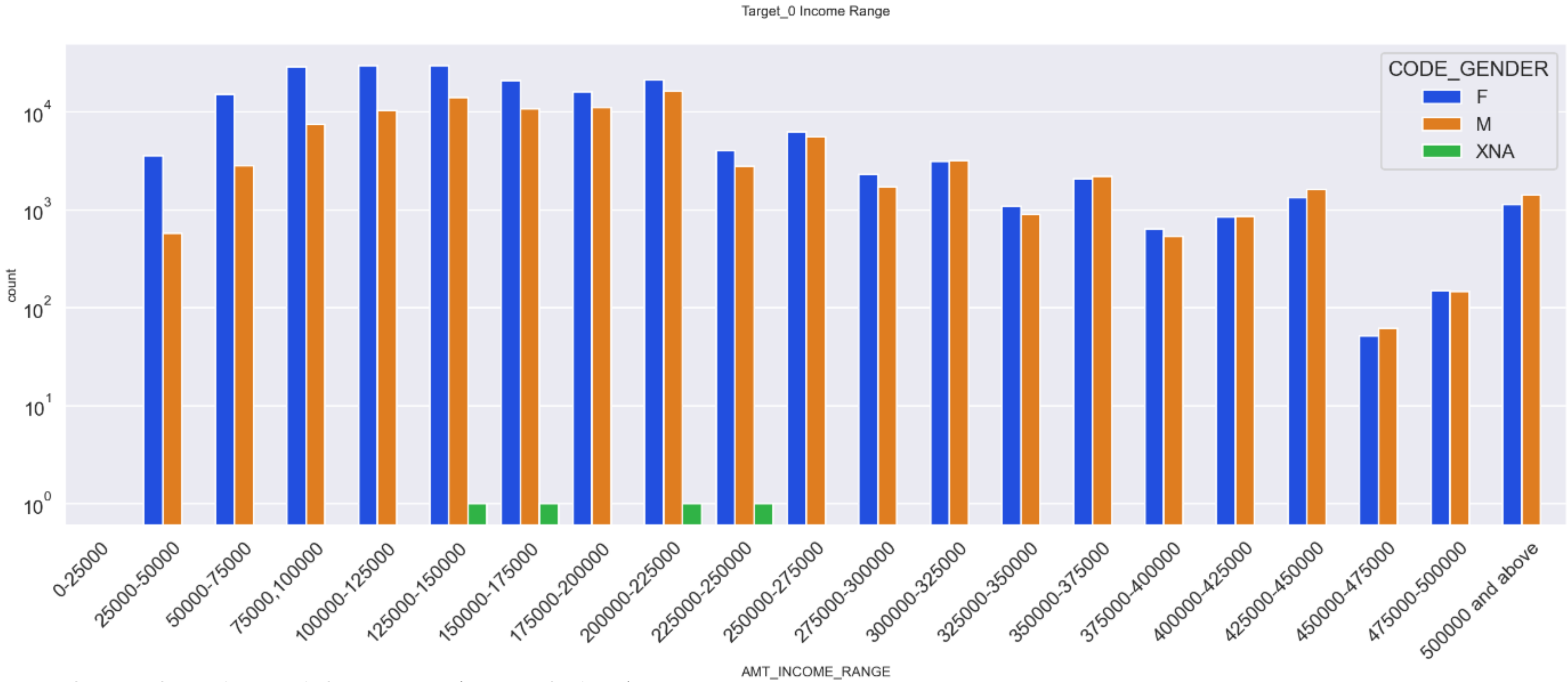


Analysis of the Code : Gender
CODE_GENDER
XNA
M
F
0.0
0.1
0.2
0.3
0.4
0.5
0.6



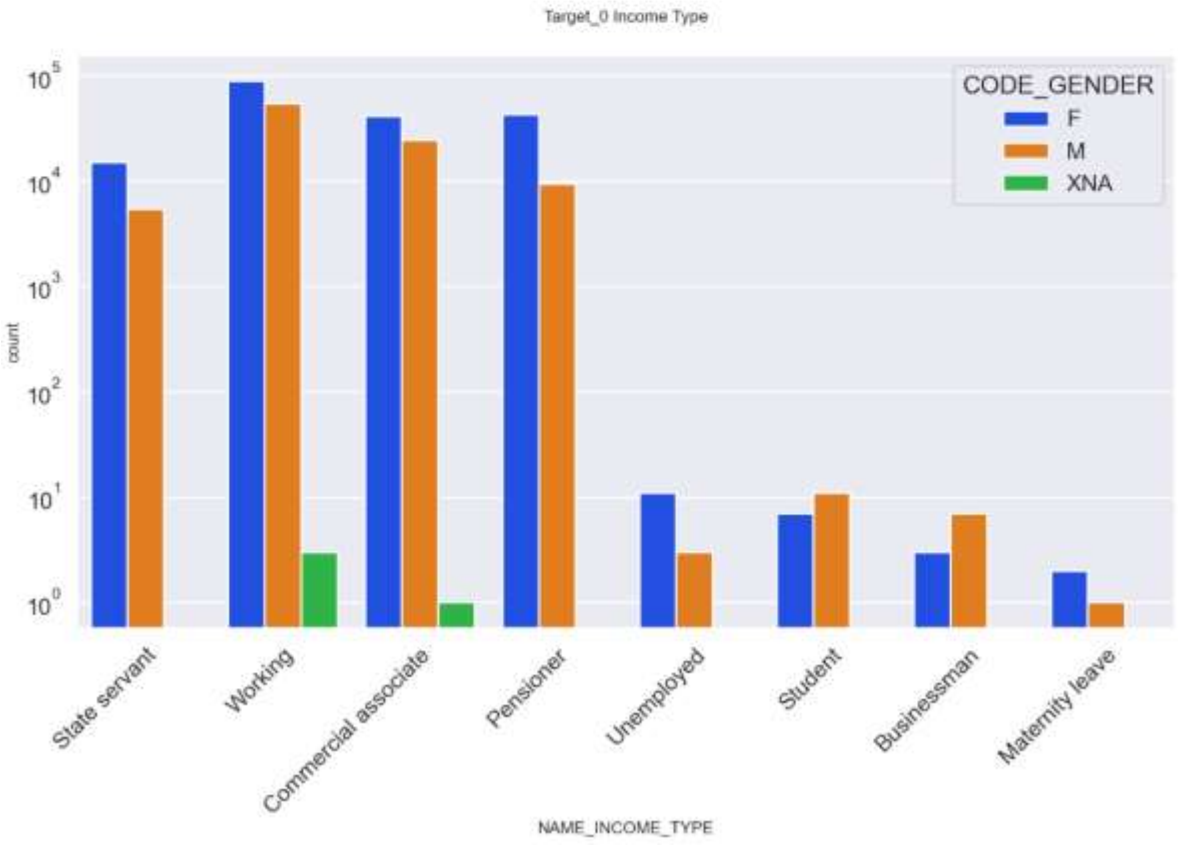
The analysis of AMT_GOODS_PRICE
AMT_GOODS_PRICE
10^6
10^5

Univariate Analysis



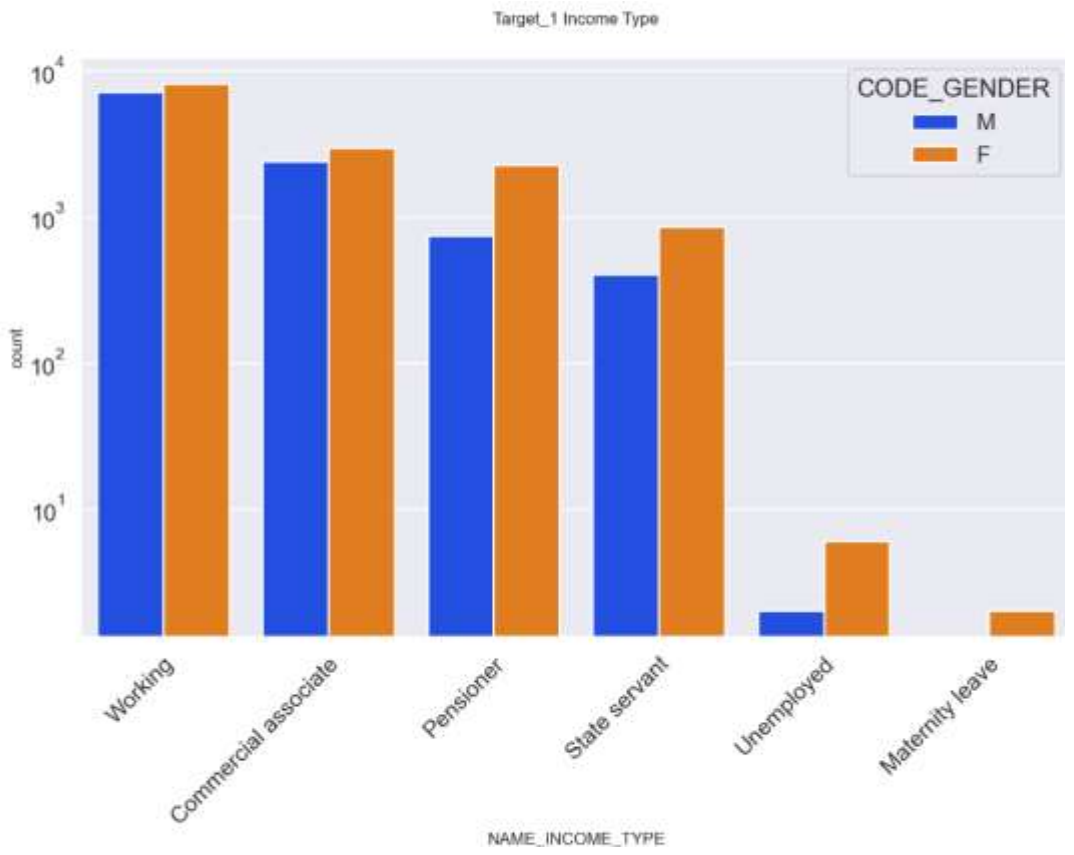
Inference from the graph for target=0 (Non-Defaulters) are:
The number of females is greater than that of males.
The data indicates that females have more credits in the specified range compared to males.

Univariate Analysis



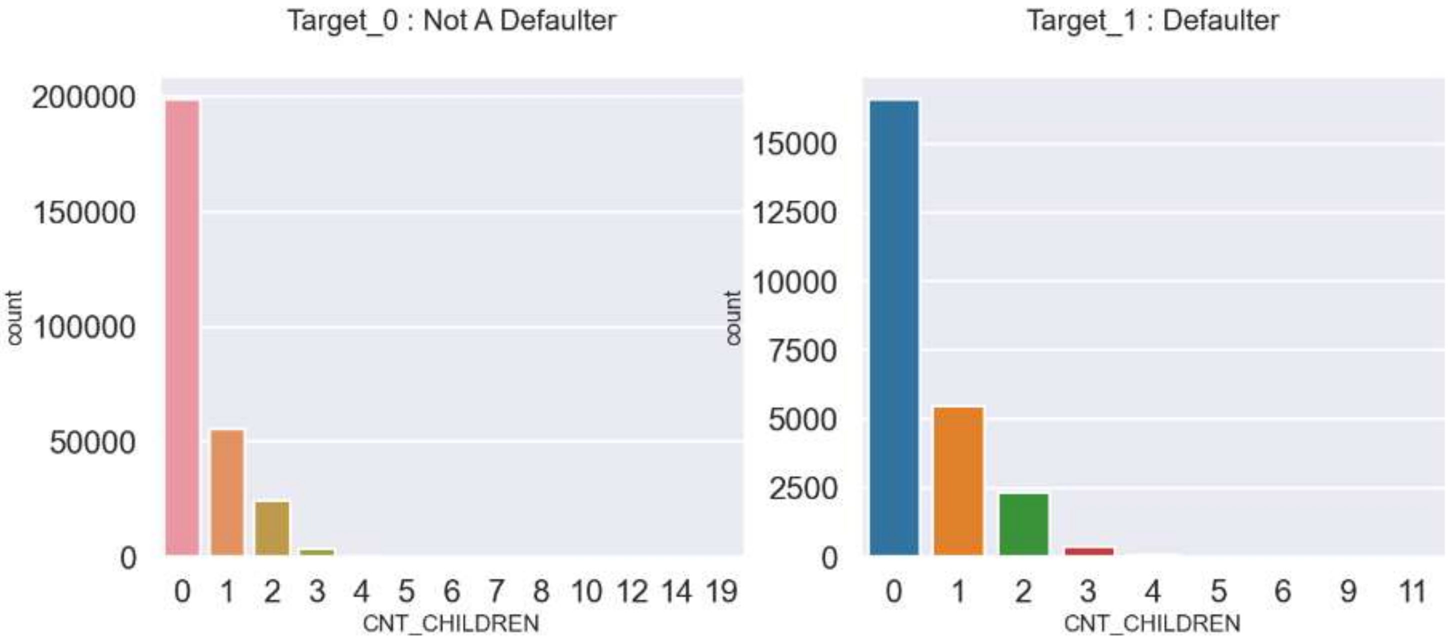
Inference:
Key observations from the graph for Target_0 (Non-Defaulters) are:
Females have higher credit counts compared to males.
The highest credit counts are associated with income types such as working, commercial associate, pensioner, and state servant.
The lowest credit counts are associated with income types such as student, unemployed, businessman, and maternity leave.

Univariate Analysis



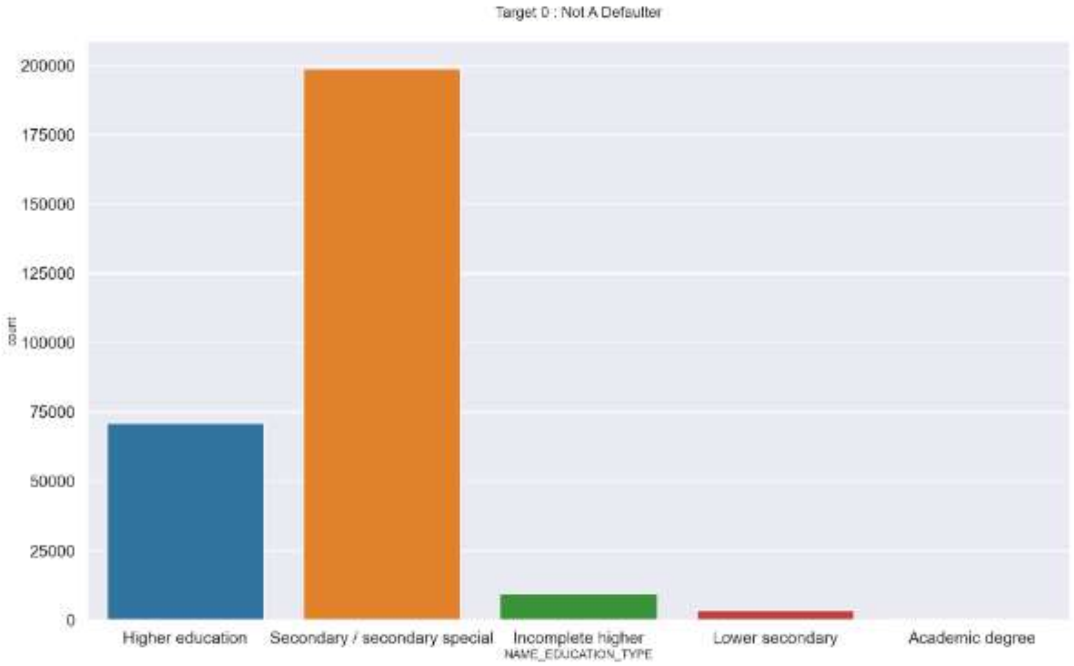
Inference:
Key observations from the graph for Target=1 (Defaulters) are:
The highest credit counts are seen for income types such as working, commercial associate, pensioner, and state servant, similar to Target=0.
The lowest credit counts are observed for income types such as unemployed and maternity leave.

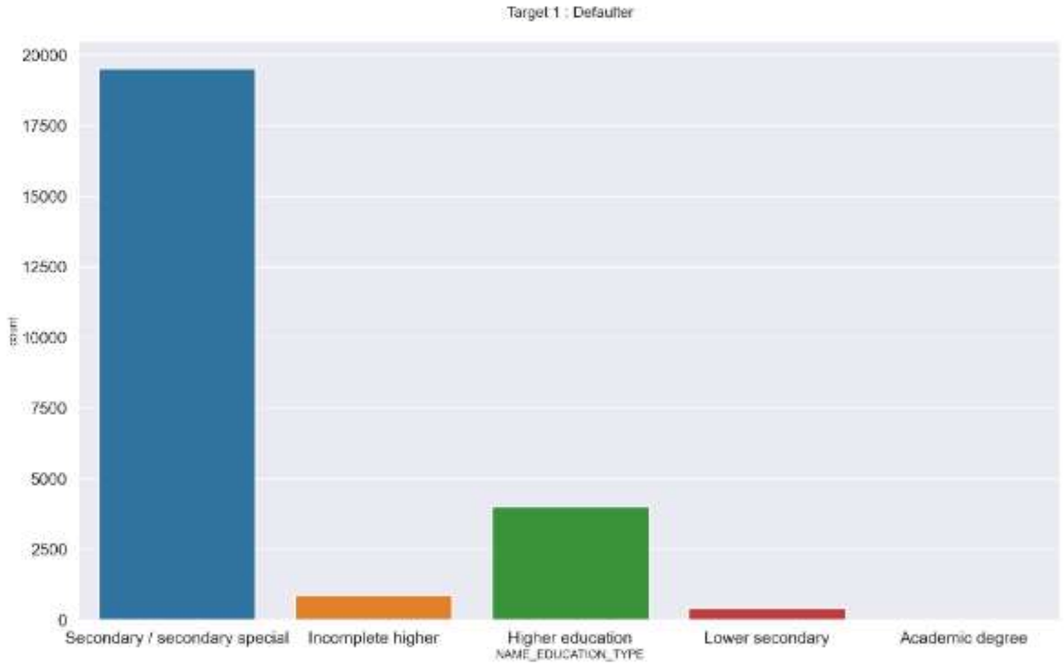
Univariate Analysis



Inference: Key takeaways from the graph:
A low number of children is associated with both higher chances of being a defaulter and non-defaulter.
Therefore, no specific conclusions can be drawn from this data exploration.

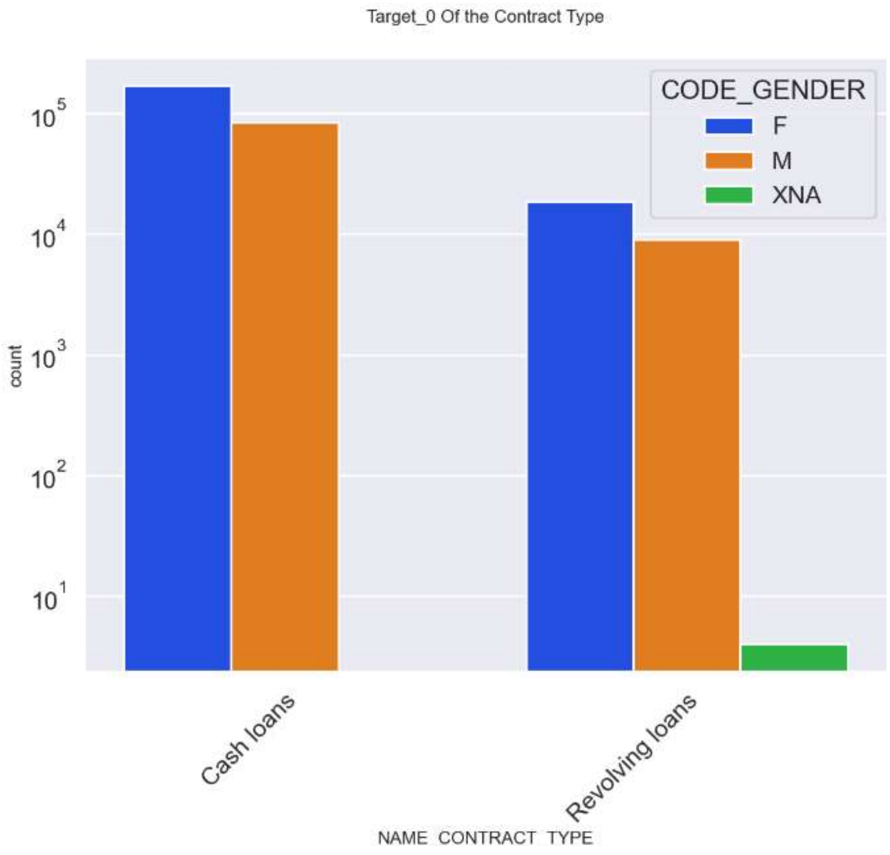
Univariate Analysis





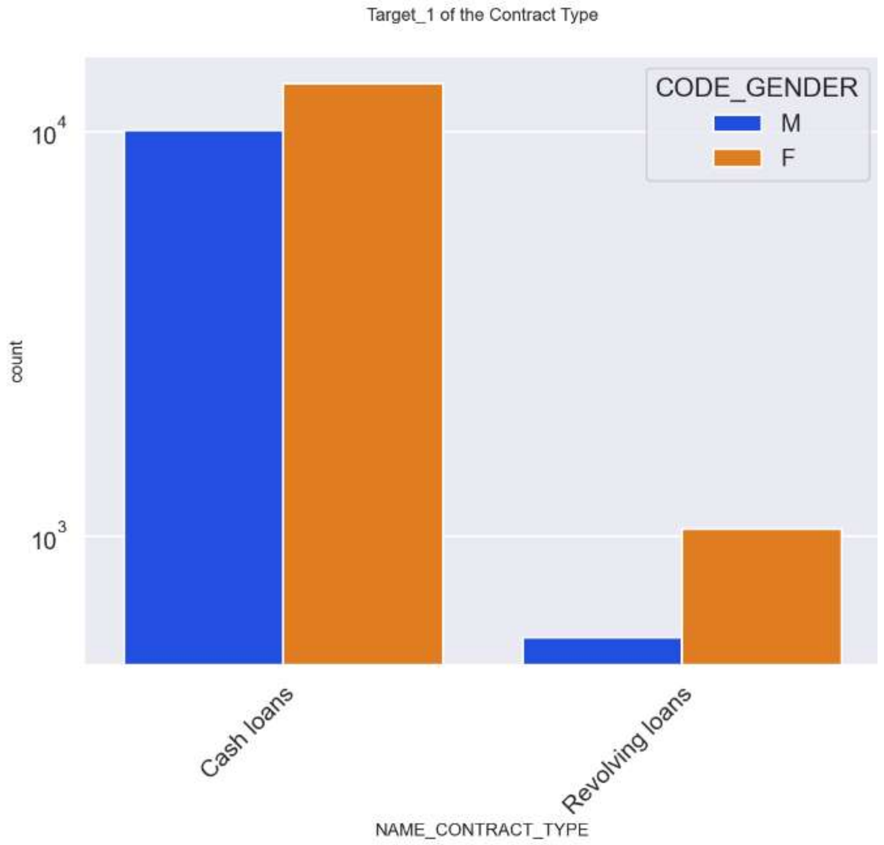
Inference: Key observations from the graph:
The comparison reveals that individuals with secondary education have the highest rate of defaulting.

Univariate Analysis



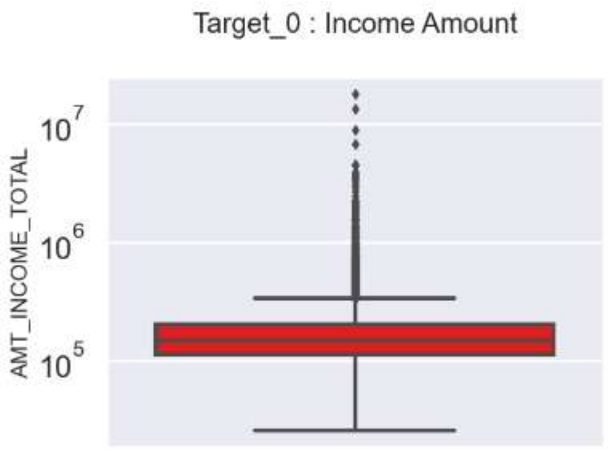
Inference: Key takeaways from the graph for Target=0 (Non-Defaulters):
Cash loan contracts have a higher number of credits compared to revolving loan contracts.
The count of females is higher.

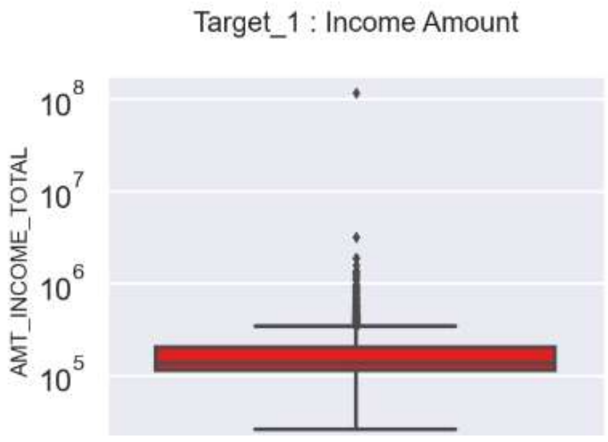
Univariate Analysis



Inference: Key takeaways from the graph for Target = 1 (Defaulters):
Cash loan contracts have a higher number of credits compared to revolving loan contracts.
Revolving loans are exclusively held by females.

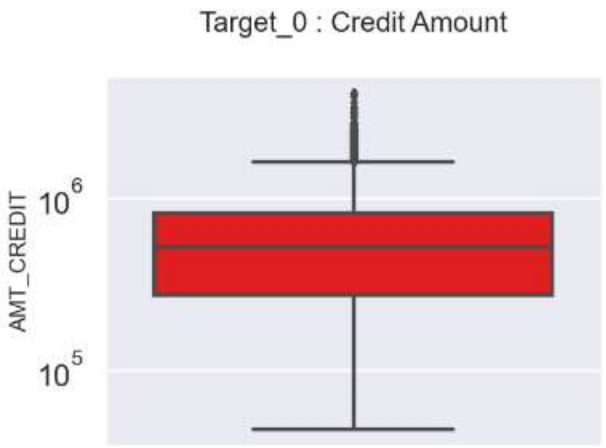
Univariate Analysis

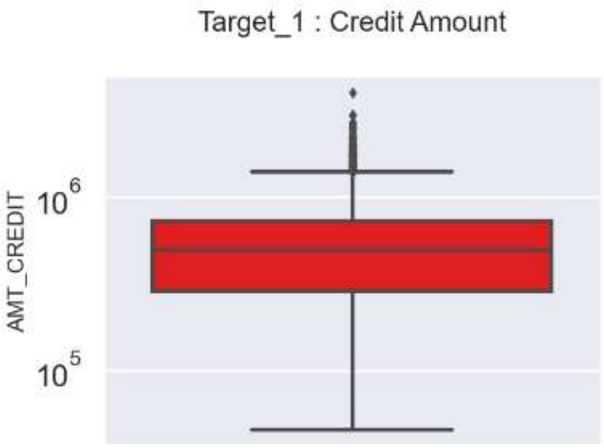




Inference: Key observations from the two graphs:
Outliers are present in both datasets.
The third quartile is relatively narrow for both Target 1 and Target 0.
The majority of clients have income levels in the first quartile.

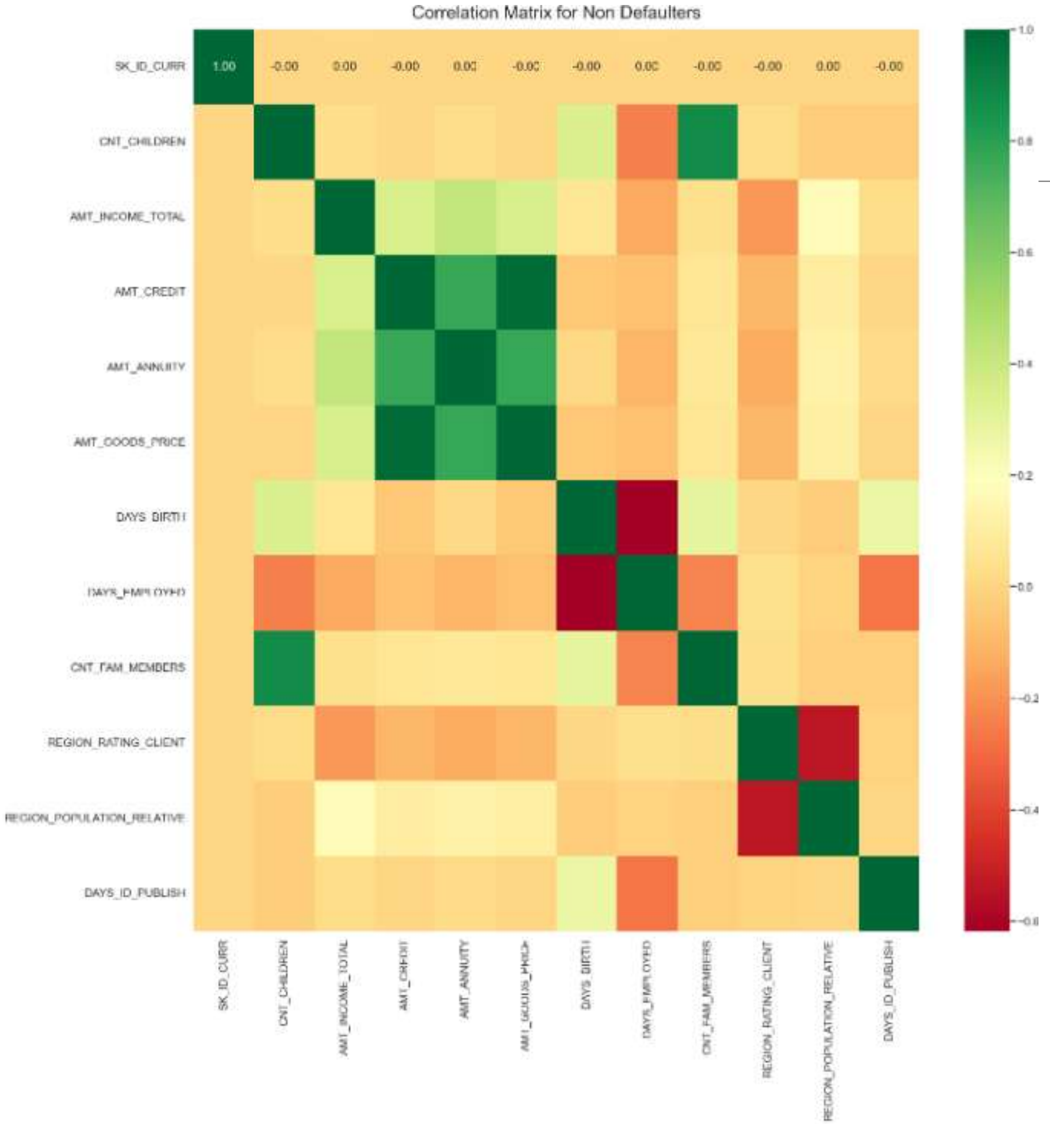
Univariate Analysis:

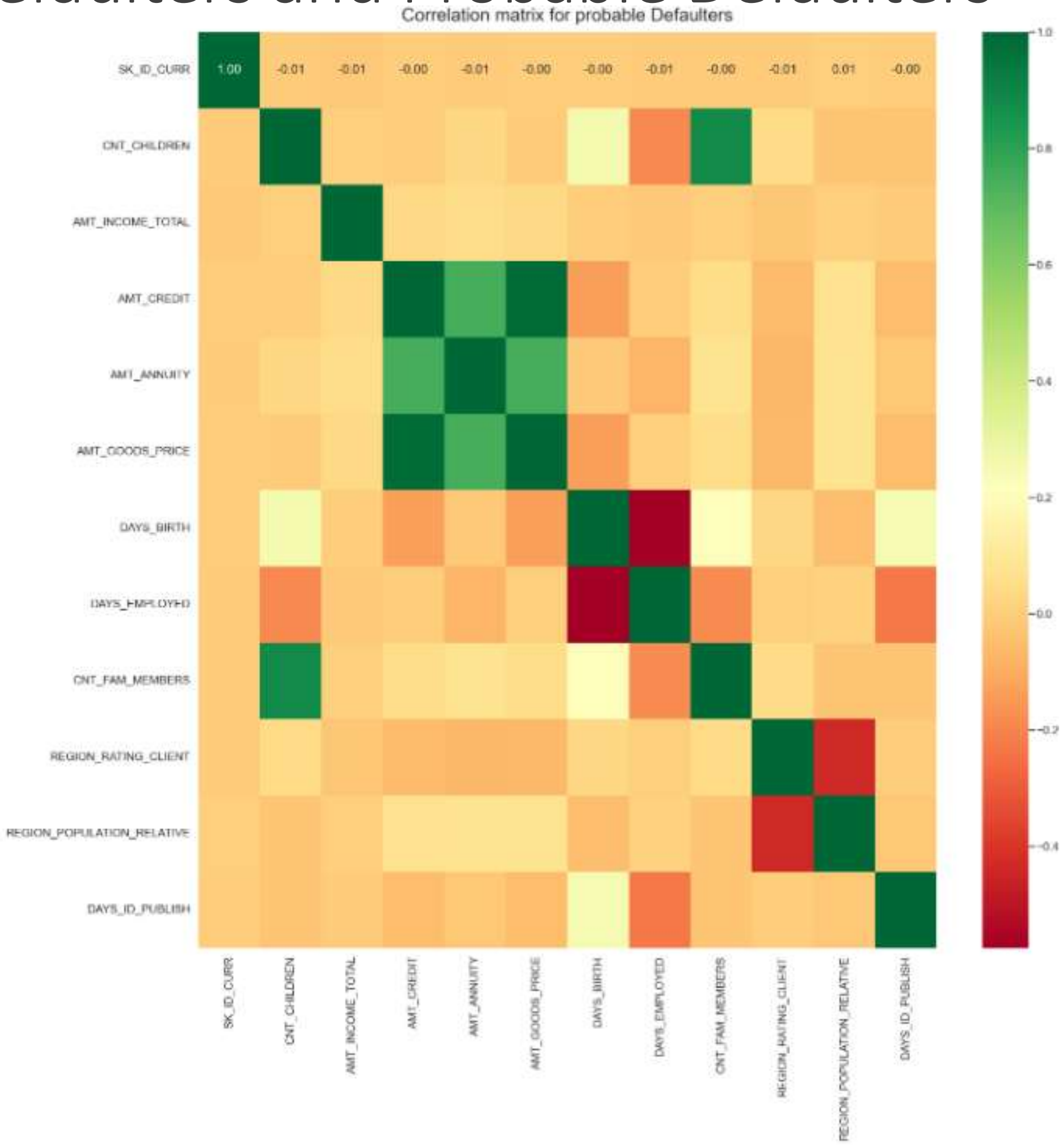




Inference:
Key takeaways from the two graphs:
Outliers are evident in both datasets.
The third quartile is narrow for both Target 1 and Target 0.
The majority of clients have credit amounts in the first quartile.

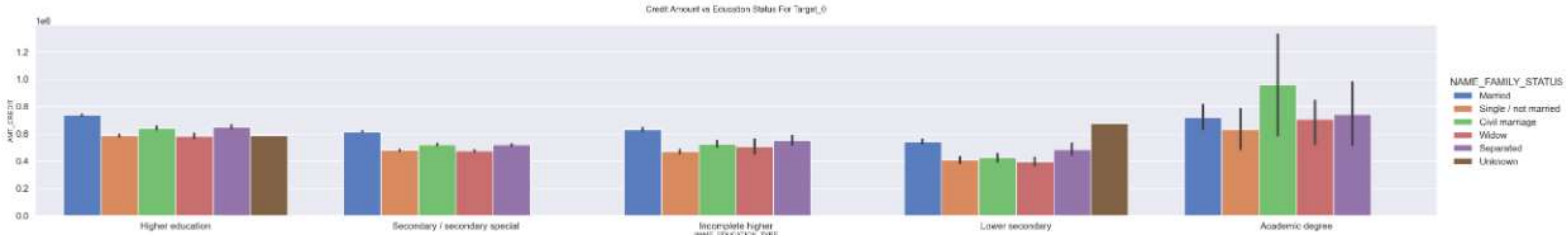
Bivariate Analysis: Correlation for Non Defaulters and Probable Defaulters





Credit amount to Education status

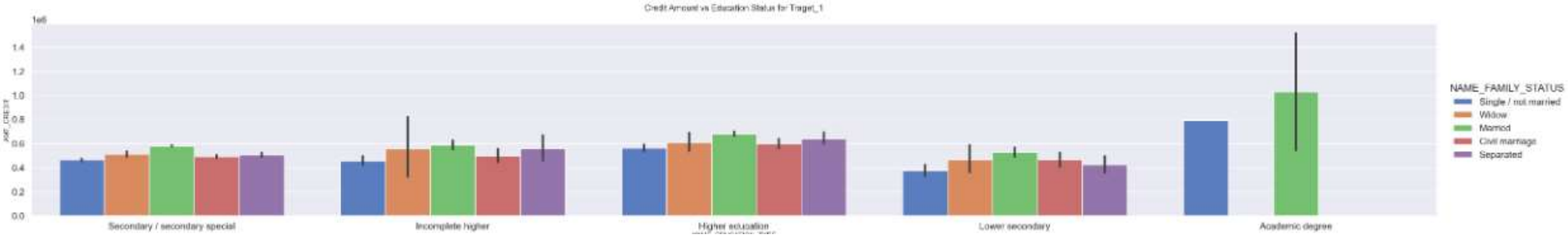
Plotting of the NAME_EDUCATION_TYPE vs AMT_CREDIT for Target_0 for each family status



Inference: Key observations from the above graph for target = 0 (Non-Defaulters):
Customers with an academic degree have higher credit amounts, with the civil marriage segment having the highest credit.
Lower educated customers tend to have lower credit amounts, with widows having the lowest.

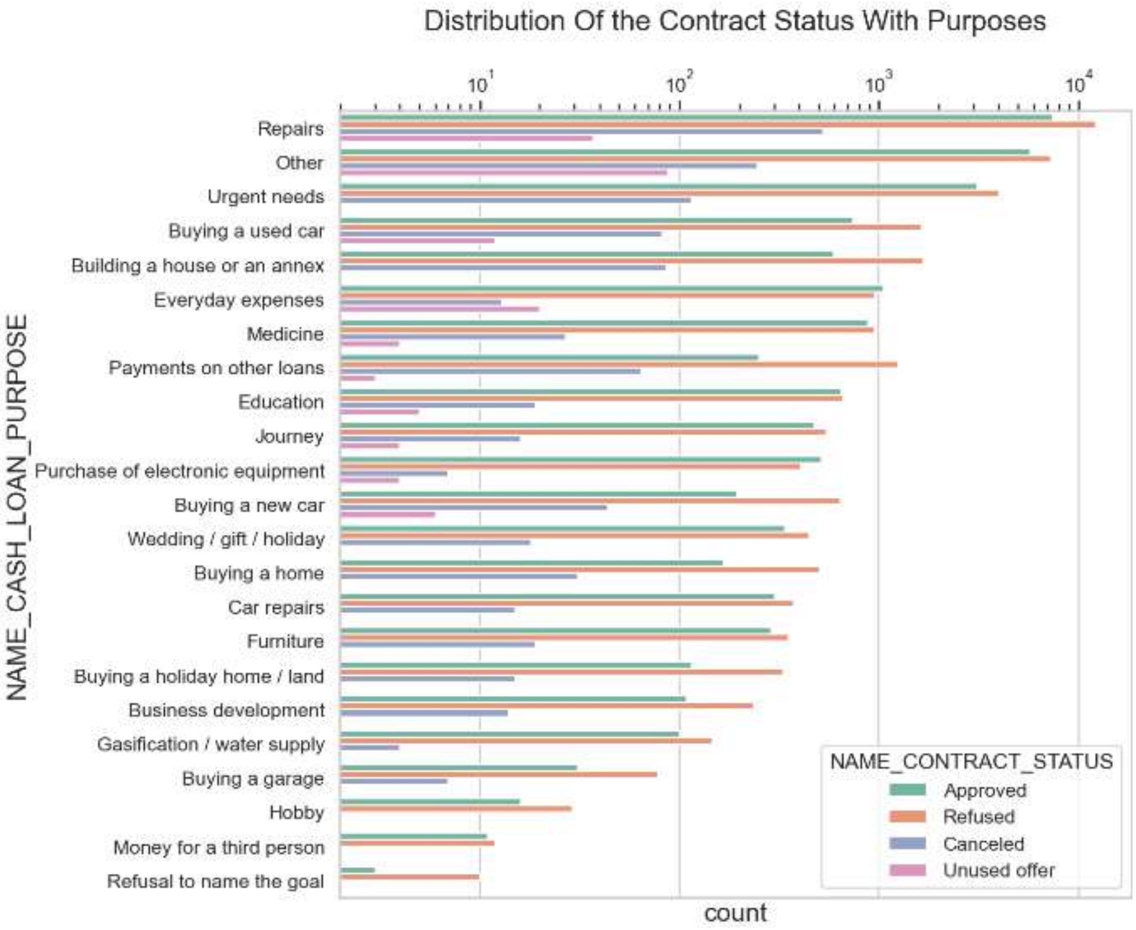
Credit amount to Education status

Plotting of the NAME_EDUCATION_TYPE vs AMT_CREDIT for Target_1 for each family status



Inference: Key observations from the above graph for target = 1 (Defaulters):
Married customers with academic degrees generally have higher credit amounts, which corresponds to a higher default rate.
Across all education segments, married customers tend to have higher credit amounts.
Customers with lower education levels tend to have lower credit amounts.
Single and married are the only two family types present among those with academic degrees.

Univariate analysis with the Merged data



Key takeaway - The majority of loan rejections were for the purpose of 'Repairs'

Bivariate analysis with the Merged data

In this Box Plot for Previous credit vs Housing type, we can conclude that the bank should avoid giving loans for co-op apartments, as these applicants are experiencing difficulties with payments.

Thank you